MILITARY SPECIFICATION

BOMB, LOADING PRACTICE BDU-15/E
BOMB, LOADING PRACTICE BDU-16/E

1. SCOPE. This specification covers the requirements for two types of dummy bombs designated as BDU-15/E and BDU-16/E, respectively. Both dummy bombs are for use in crew training in handling and loading in respective aircraft. (See 6.1)

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-T-266	Thinner; Dope and Lacquer
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-B-131	Barrier Material; Waterproof, Flexible
MIL-C-132	Crates; Open Wood, Maximum Capacity 2500 Pounds
MIL-V-3083	Vibrator, Concrete, Pneumatic
MIL-L-7178	Lacquer, Cellulose Nitrate, Gloss, For Aircraft Use
MIL-T-9107	Test Reports, Preparation Of
MIL-B-27000	Bombs, Practice, General Specification For
MIL-D-70327	Drawings, Engineering and Associated List

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

FSC 1115

AF Hooven-Dayton Co., 24 May 1962 1400

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DRAWINGS

Air Force

61J14325	Bomb, Loading Practice BDU-15/E
61J14327	Nose, Bomb, Loading Practice BDU-15/E, Assembly
61J14328	Warhead, Bomb, Loading Practice BDU-15/E, Assembly
61J14329	Afterbody, Bomb, Loading Practice BDU-15/E, Assembly
61D14456	Bomb, Loading Practice BDU-16/E
61D14472	Conversion Kit, BDU-16/E Practice Loading Bomb KMU-157/E

(Copies of documents required by contractors in connection wit' specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.2 COMPONENTS

3.2.1 BOMB, LOADING PRACTICE BDU-15/E. The BDU-15/E bomb shall consist of the following major components:

	Item Name	Drawing Number	Quantity
a.	Nose, Bomb, Loading Practice BDU-15/E	61J14327	1
b.	Warhead, Bomb, Loading Practice BDU-15/E	61J14328	1
c.	Afterbody, Bomb, Loading Practice BDU-15/E	61J14329	1

3.2.2 BOMB, LOADING PRACTICE BDU-16/E. The BDU-16/E bomb shall consist of the following major components:

	Item Name	Drawing Number	Quantity
a.	Warhead, Bomb, Loading Practice BDU-15/E	61J14328	1

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b. Afterbody, Bomb, Loading Practice BDU-15/E	61J14329	1
c. Conversion Kit, BDU-16/E Practice Loading Bomb KMU-157/E	61D14472	1

3.3 GENERAL SPECIFICATION. The requirements of MIL-B-27000 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 DESIGN AND CONSTRUCTION

3.4.1 BOMB, LOADING PRACTICE BDU-15/E. The BDU-15/E Bomb shall be designed and constructed in conformance with drawing 61J14325 and subsidiary drawings.

3.4.2 BOMB, LOADING PRACTICE BDU-16/E. The BDU-16/E Bomb shall be designed and constructed in conformance to drawing 61D14456 and subsidiary drawings.

3.4.2.1 CONVERSION KIT, BDU-16/E PRACTICE LOADING BOMB KMU-157/E. The KMU-157/E Conversion Kit shall conform to drawing 61D14472 and subsidiary drawings.

3.4.3 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria listed in MIL-D-70327 shall apply and new drawings shall be prepared as required.

3.4.4 CONCRETE FILL. The concrete fill process shall be accomplished by the contractor on all bombs.

3.4.4.1 DENSITY. The density of the concrete fill shall be 143 pounds per cubic foot, plus 7, minus 18 lb/ft^3.

3.4.4.2 MIXING PROCEDURE. The recommended procedure for mixing is as specified in MIL-B-27000.

3.4.4.3 METHOD OF FILLING. With the warhead section in a vertical position with aft end down fill with concrete by performing the following steps:

a. Fill with concrete to 1/4 the distance to the fill line as shown on drawing 61J14328.

b. Insert vibrator conforming to MIL-V-3083 and vibrate for a period of 30 to 60 seconds and remove vibrator.

c. Fill to 1/2 the distance to the fill mark.

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d. Repeat step b.

e. Fill to 3/4 the distance to the fill mark.

f. Repeat step b.

g. Fill to the fill mark.

h. Repeat step b.

3.4.4.4 CURING TIME. Curing time for the concrete fill shall be as specified in MIL-B-27000.

3.4.5 WEIGHT. Weight of the practice bombs shall be as follows:

a. BDU-15/E, 2244 pounds ± 50 pounds.

b. BDU-16/E, 2320 pounds ± 50 pounds.

3.4.6 CENTER OF GRAVITY. Center of gravity for the practice bombs shall be as follows:

a. BDU-15/E, Station 57.25, plus 2.25 inch, minus 0.50 inch.

b. BDU-16/E, Station 69.0 ± 0.5 inch.

3.4.7 STATIC LOADS. The practice bombs shall be capable of withstanding the loads imposed by transportation, handling and flight in carrier aircraft.

3.5 FINISH. Finish process shall be in accordance with MIL-B-27000.

3.5.1 FINAL FINISH COAT. The final finish coat shall be applied to external surfaces as follows:

a. One coat of Lacquer, Cellulose Nitrate, Gloss, color aluminum in accordance with MIL-L-7178.

b. Two coats of Lacquer, Cellulose Nitrate, Gloss, clear in accordance with MIL-L-7178. The lacquer shall be thinned by one part of lacquer thinner conforming to TT-T-266 to two parts of lacquer. Each coat shall be applied by spraying on a coat 0.5 to 1.0 mil in thickness. Parts shall be allowed to air dry for a minimum of one hour after receiving the finish coat, prior to handling.

3.6 PERFORMANCE. Satisfactory performance shall consist of proper simulation of assembly and loading and simulation of electrical checks of the parent weapon.

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3.7 ENVIRONMENTAL REQUIREMENTS. The BDU-15/E and BDU-16/E Bombs shall be capable of satisfactory performance after subjection to the following environmental conditions:

a. Vibration likely to be encountered in transportation, and flight.

b. Shock likely to be experienced in transportation, handling, loading and flight.

3.8 IDENTIFICATION OF PRODUCT. The bombs, assemblies and major parts shall be marked for identification in accordance with MIL-STD-130.

3.9 WORKMANSHIP. Workmanship shall be in accordance with the requirements of MIL-B-27000.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. Inspection and testing of the practice bombs shall be classified as follows:

a. Preproduction tests.

b. Acceptance tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test units shall be representative of the production practice bombs, complete with concrete fill. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's facility under supervision of the procuring activity.

EXCEPTION: The fit test shall be conducted at an Air Force facility.

4.2.2 PREPRODUCTION TEST REPORTS. In the event that preproduction testing is conducted at the contractor's facility, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report in accordance with MIL-T-9107 and furnish three complete copies to the procuring activity.

4.2.3 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described in "4.5 TEST METHODS."

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of the following individual tests:

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a. Examination of Product.

b. Conformance to Drawings.

4.4 TEST CONDITIONS. Test conditions shall be as specified for each specific test in "4.5 TEST METHODS."

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each bomb shall be given a complete visual examination to determine compliance with all requirements of this specification for which no tests are specified. Failure to satisfy these requirements shall be cause for rejection.

4.5.2 CONFORMANCE TO DRAWINGS. Each bomb shall be inspected for strict conformance to the engineering drawings and all specifications and standards referenced therein. Failure to pass this inspection shall be cause for rejection.

4.5.3 WEIGHT TEST. The practice bombs, filled with concrete and fully assembled shall be weighed using a commercial crane scale or equivalent. The weights shall be as listed below:

a. BDU-15/E, 2244 pounds ± 50 pounds.

b. BDU-16/E, 2320 pounds ± 50 pounds.

Failure of weight to fall within the above limits shall be cause for rejection.

4.5.4 CENTER OF GRAVITY. Location of the center of gravity of the bombs, after concrete filling and complete assembly, shall be determined in any convenient manner. Centers of gravity shall be as listed below:

a. BDU-15/E, Station 57.25, plus 2.25 inch, minus 0.50 inch.

b. BDU-16/E, Station 69.00 ± 0.5 inch.

Failure of the cg location to fall within the above limits shall be cause for rejection.

4.5.5 STATIC LOAD TEST

4.5.5.1 TEST SPECIMEN PREPARATION. Prior to performing the static tests, the applicable bomb shall be filled with concrete as specified and fully assembled.

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4.5.5.2 TEST SET UP. Test set up shall be in accordance with Procedure I, Suspension Test of MIL-B-27000 except that 30-inch suspension system and sway brace simulation shall not apply.

4.5.5.3 TEST PROCEDURE. Test procedure shall be as specified in Procedure I, Suspension Test of MIL-B-27000. Design limit loads are as follows:

a. Longitudinal; 11600.00 pounds.

b. Vertical, positive and negative; 11600.00 pounds.

c. Lateral, 11600.00 pounds.

4.5.6 CONTINUITY TEST. Using any suitable instrumentation, check each circuit shown in figure 1 and figure 2 for continuity. Indication of "no continuity" in any circuit shall be cause for rejection.

4.5.7 VIBRATION TEST. The practice bomb shall be subjected to a vibration test as described in MIL-B-27000. Subsequent to the vibration test, the Continuity Test shall be performed. Evidence of damage, failure of any part or failure to pass the continuity test shall be considered as failure to pass the vibration test and cause for rejection.

4.5.8 SHOCK TEST. The practice bombs, with specified concrete fill and fully assembled shall be subjected to the shock test specified in MIL-B-27000 with the following exceptions:

a. 30 inch suspension system and sway brace simulation shall not apply.

b. Shock intensities shall be as follows:

(1) Along horizontal axis, 5g $\pm$ 10 percent.

(2) Along vertical axis, 5g $\pm$ 10 percent.

(3) Along lateral axis, 5g $\pm$ 10 percent.

Subsequent to the shock test, the continuity test shall be performed. Damage to any part, or failure to pass the continuity test shall be considered as failure to pass the shock test and cause for rejection.

4.5.9 FIT TEST. The practice bombs shall be subjected to a fit test to determine compatibility with applicable aircraft and parent weapon handling and loading equipment.

NOTE: This test shall be conducted at an Air Force facility.

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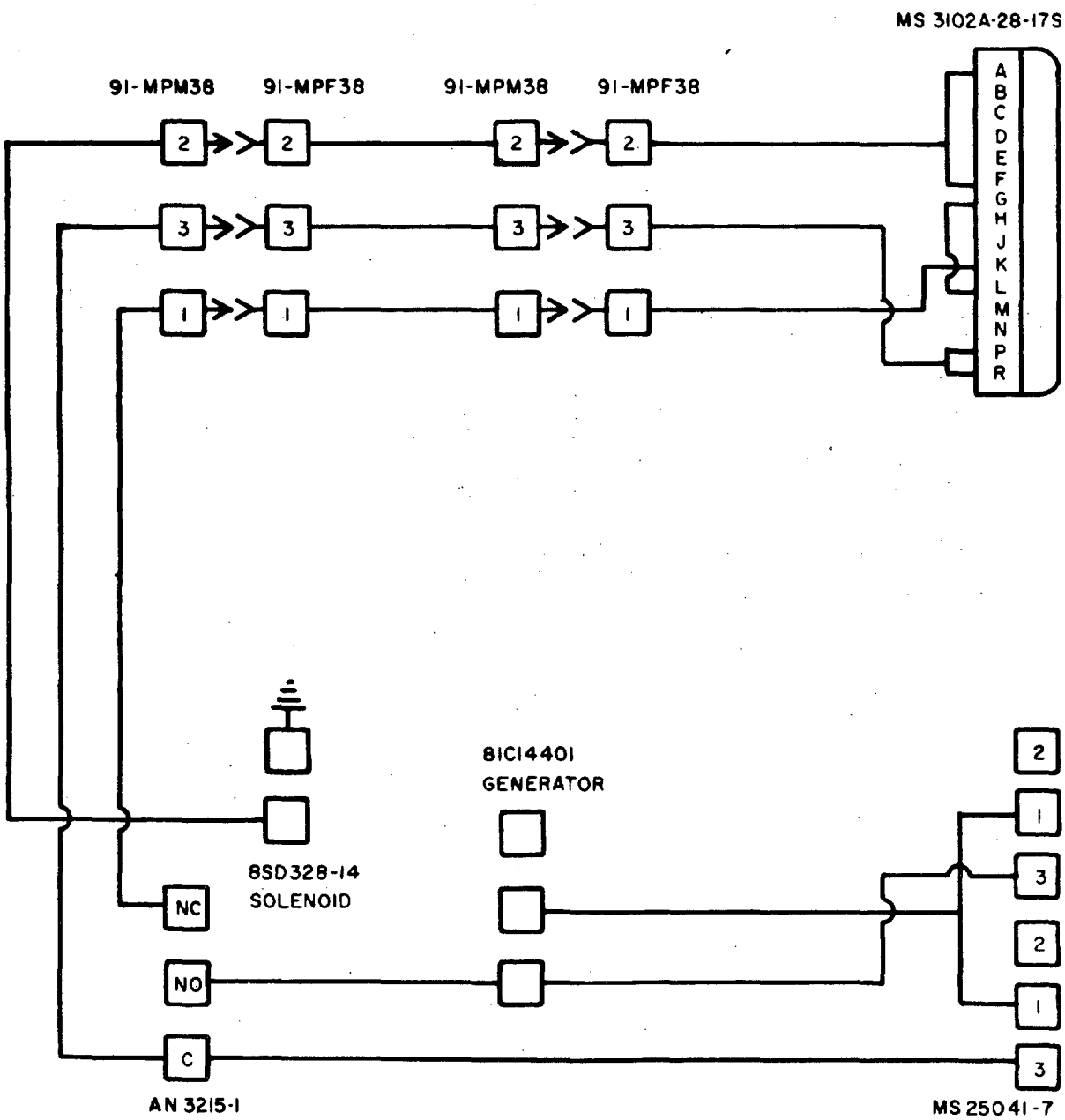


FIGURE 1 BDU-15/E WIRING DIAGRAM

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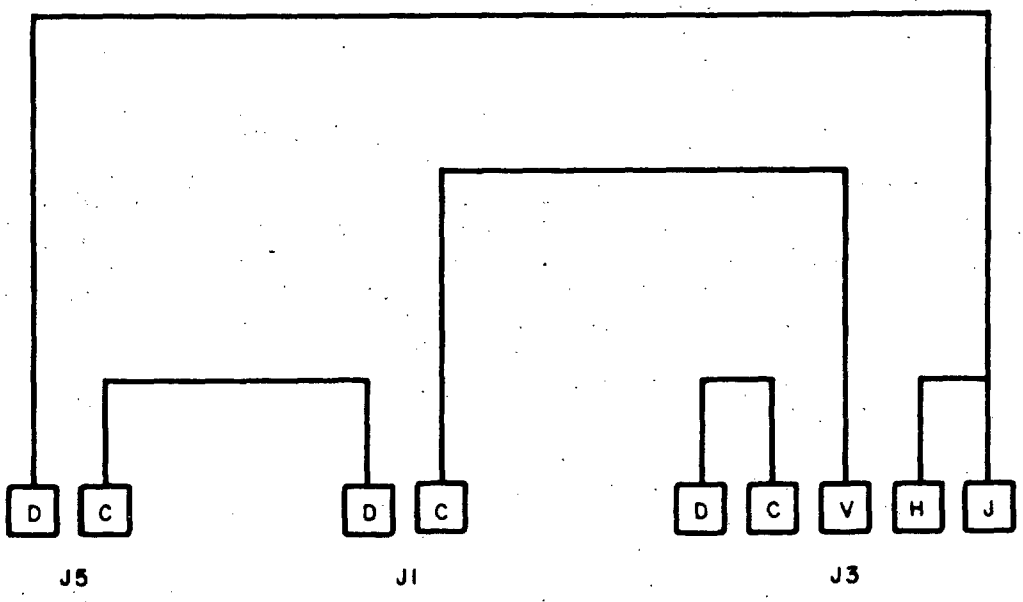


FIGURE 2 WARHEAD WIRING DIAGRAM

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5. PREPARATION FOR DELIVERY

5.1 PRESERVATION, ALL LEVELS, BOMB ASSEMBLED OR DISASSEMBLED. All machined surfaces not having a protective finish shall be preserved in accordance with Method I, MIL-P-116. Disassembled bombs and assembled bombs with or without the parachute, shall have all openings covered with barrier material conforming to MIL-B-131, and sealed with tape conforming to PPP-T-60. Openings in the bomb include, but are not limited to the following:

a. Screw holes without screws.

b. Openings for safing pins.

c. Peripheral separations.

d. All large openings when bomb is shipped disassembled.

5.2 PACKAGING AND PACKING, ALL LEVELS. The bomb or components as designated by the procuring activity shall be packed in a crate conforming to MIL-C-132. The fins and attaching hardware shall be removed from the bomb and secured in the crate. Physical protection shall be in accordance with JAN-P-100.

5.2.1 FOR AIR SHIPMENT. The bomb shall be completely assembled and mounted on a MHU-47/E Trailer, Dummy Bomb.

5.3 MARKING. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be Bomb, Loading Practice BDU-15/E.

6. NOTES

6.1 INTENDED USE. The Bomb, Loading Practice BDU-15/E and Bomb, Loading Practice BDU-16/E are intended for use in training ground crews in handling and loading of parent weapon. These items are non-expendable and are not normally dropped from carrying aircraft.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests except fit test will be conducted by the procuring activity.

c. Levels of packaging and packing.

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6.2.1 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that two units representative of the production BDU-15/E practice bomb or BDU-16/E practice bomb, as applicable, will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.3 PRODUCTION DELIVERY. No production items should be delivered prior to approval of the preproduction samples.

6.4 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative models of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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